Individual Case Study: Maryland

Maryland was selected for an individual case study in order to examine recent measles outbreaks. Data was taken from Project Tycho (historic trends), the Maryland Department of Health (confirmed measles cases), and SchoolVaxView (MMR coverage) (Van Panhuis et al., 2018; Center for Immunization, 2026; 2026). The data was analyzed through Google Sheets.

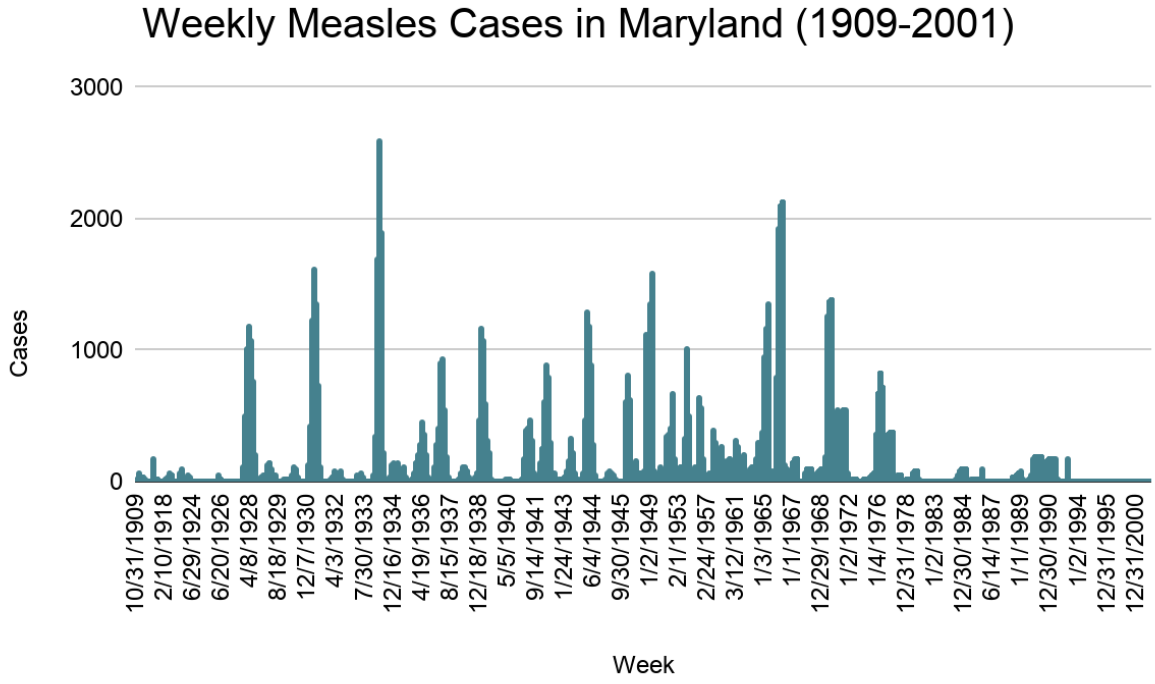


Figure 1. Weekly Measles Cases in Maryland from 1909 to 2001. The number of cases per week period is taken from Project Tycho for measles cases in Maryland from 1909 to 2001 (Van Panhuis et al., 2018). An assumption made for this data is that the time period is inclusive, as it is not specified whether cases occurring on the end date are included. The highest number of cases recorded in a week was 2,597 between April 29, 1934 and May 5, 1934. The lowest number of cases recorded in a week was 0 for 1,744 weeks within the dataset.

Viewing cases in a week period limits visualization; therefore, using a pivot table, the week start date was used to create a sum for an annual case overview.

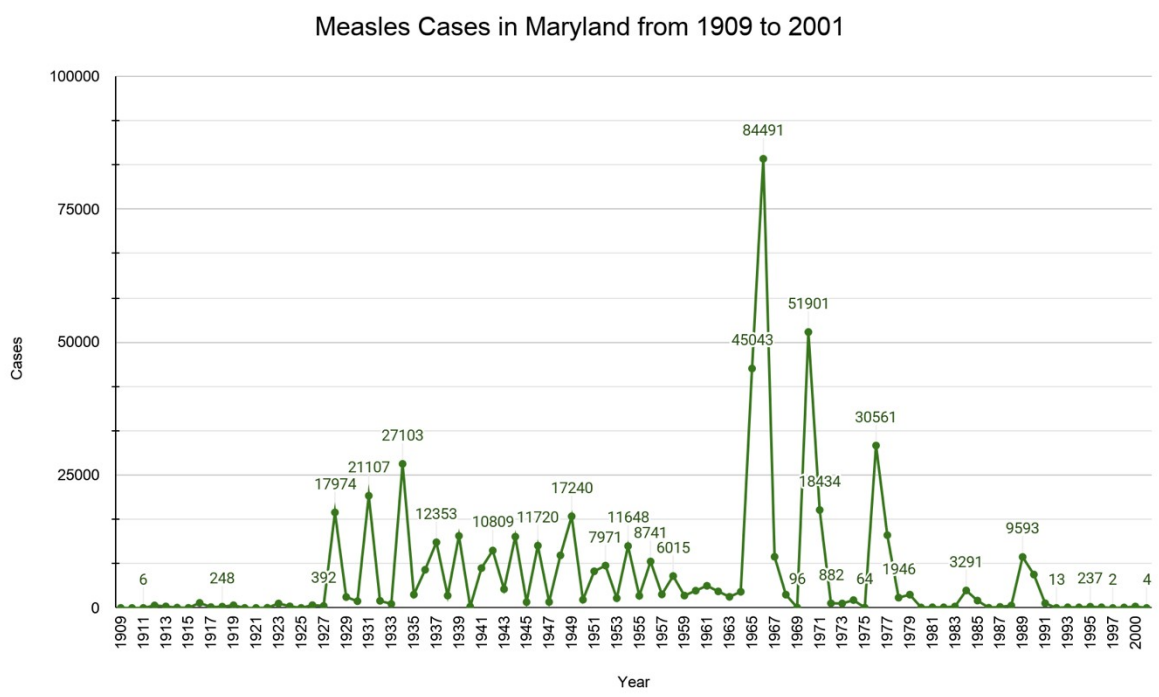


Figure 2. Annual Measles Cases in Maryland from 1909 to 2001. This data uses the start of the week as the time to aggregate by year the sum of the measles cases from 1909 to 2001. Toward the end of the 20th century, the number of measles cases declines in comparison to the mid-1900s. The data source is Project Tycho (Van Panhuis et al., 2018). The highest number of cases recorded in a year was 84,491 in 1966 while the lowest number of cases occurred in 1910.

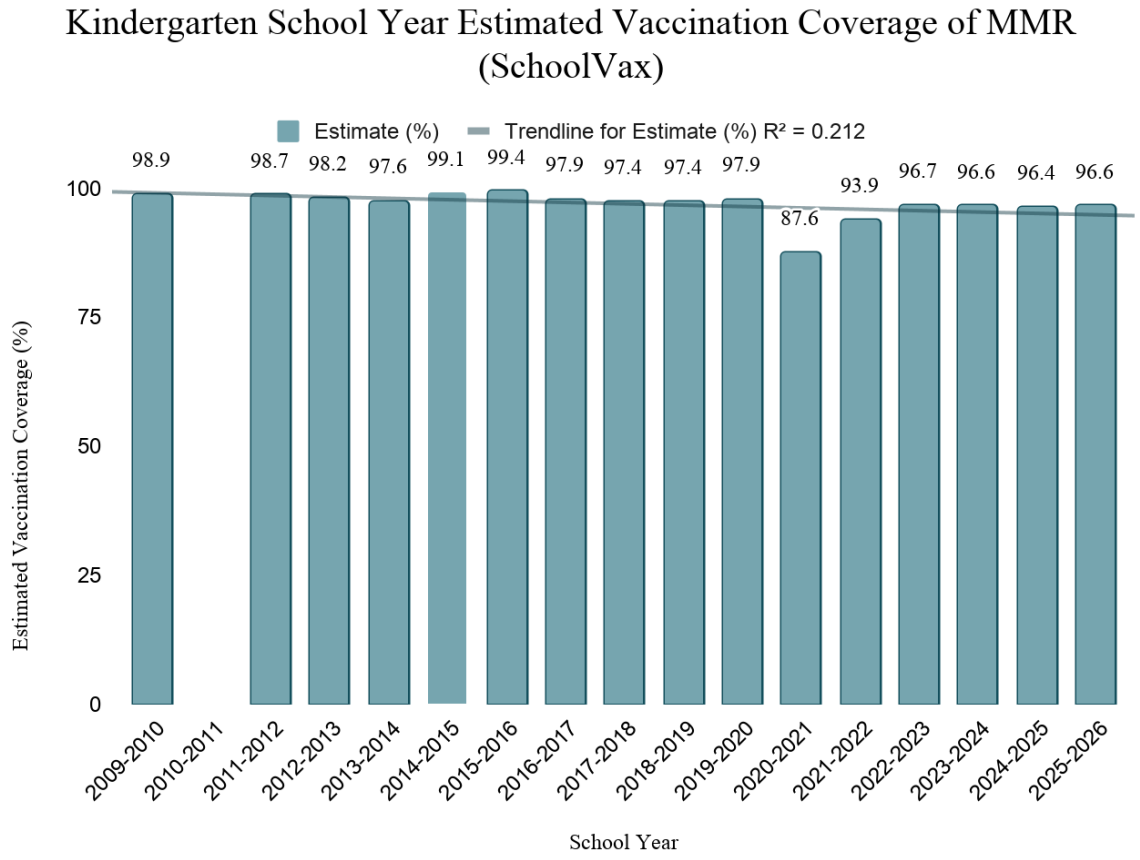


Figure 3. School Year Estimated Vaccination Coverage of MMR in Kindergartens from 2009-2010 school year to the 2025-2026 school. Based on SchoolVaxView data, the estimated coverage for MMR vaccination for Maryland Kindergarteners was analyzed. The data from the 2010-2011 school years was not available from SchoolVaxView. The average MMR coverage within the 2009-2010 to 2025-2026 school year period was approximately 96.9% ± 2.8 with an average surveyed of about 91.9% ± 10.5. The large standard deviation for the average surveyed indicates a varied representation of sampling. The average size of the population sampled was approximately 69,098 ± 6,212. The lowest coverage occurred in the 2020-2021 school year with 87.6% coverage. The highest coverage was 99.4% in the 2015-

2016 school year. The trendline has a slight decline, indicating that the vaccination coverage has been very gradually decreasing over the past few years for kindergartners in Maryland.

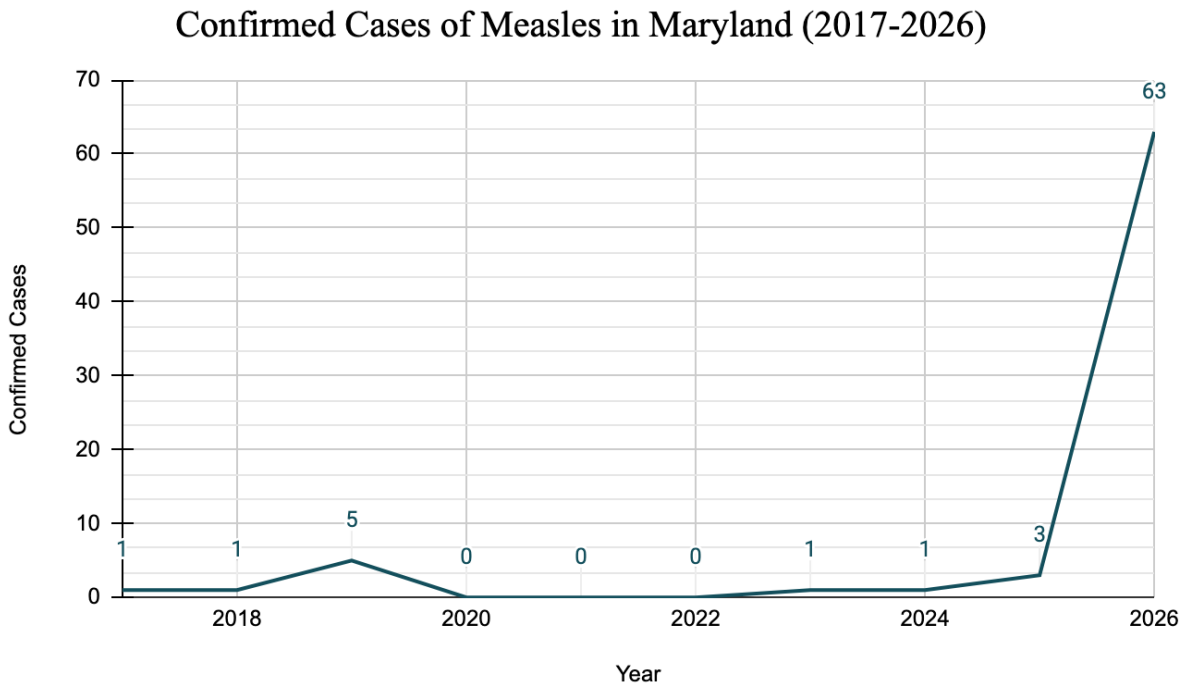


Figure 4. Confirmed Cases of Measles in Maryland from 2017 to October 23, 2025. Based on data taken from the Mayland Department of Health, cases from 2026 as of October 23 outnumber those from previous years. The number of confirmed measles cases so far in 2026 has reached 63, which is almost 6 times greater than the total cases in the past 9 years combined. This suggests present outbreaks within the state of Maryland.

While Figure 4 suggests current measles outbreaks within Maryland, the vaccination coverage of kindergartners has remained over 90% in the past 5 years. Additional research and analysis is

needed to examine whether significant relationship exist between the two or whether external factors such as travel may play a role.

References

Center for Immunization. (October 15, 2026). *Measles*. Maryland Department of Health. Accessed on October 23, 2026, from https://health.maryland.gov/phpa/OIDEOR/IMMUN/pages/measles.aspx

SchoolVaxView. (August 17, 2026). *Vaccination Coverage and Exemptions among Kindergartners.* U.S. Centers for Disease Control and Prevention. https://www.cdc.gov/schoolvaxview/data/index.html

Van Panhuis, W., Cross, A., & Bruke, D. (April 1, 2018). *Project Tycho.* MIDAS Coordination Center. 10.25337/T7/ptycho.v2.0/US.27836007